MYP MATHEMATICS 1 · SECOND EDITION

07

Lines, angles, triangles and constructions

Part IV · Shape, structure and measure

REVISION PROOF · EDITABLE SOURCE · ORIENTATION IN SPACE AND TIME

CHAPTER 7 · OPENER

Lines, angles, triangles and constructions

Angle facts, polygon structure and equal-radius constructions let us reason about shapes and check designs before building.

ENGAGEMENT

Which triples of lengths can close a triangle, and how can a workshop check a roof design before cutting timber?

Learning goals

- Use angle facts on lines, at points and with parallel lines.
- Reason with triangle and polygon angle sums.
- Apply the triangle inequality and regular-polygon facts.
- Construct perpendicular bisectors, angle bisectors and SSS triangles.

MYP lens

Key concept: Relationships

Related concepts: Space, Measurement

Global context: Orientation in space and time

ATL focus. In order for a student to think creatively and critically, students must choose an appropriate geometric fact or construction, test a design and explain why the method is reliable.

Language bank: acute · right · obtuse · reflex · vertically opposite · corresponding · alternate · co-interior · isosceles · exterior angle · polygon · perpendicular bisector · angle bisector · construction

CHAPTER 7 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

R1	Calculate 180−65 and 360−275.	READY
R2	Find 180÷3.	READY
R3	Is 95° acute, right or obtuse?	READY
R4	Find 5+6 and compare with 7.	READY
R5	Half of 6 cm is what length?	READY

Chapter map

1. **7.1 Lines, angles and angle facts** — Name the relationship first: straight line, full turn, vertically opposite or right angle.
2. **7.2 Parallel lines and triangles** — Use corresponding, alternate and co-interior angles only when the lines are parallel.
3. **7.3 Polygons and the triangle inequality** — Split polygons into triangles; test whether three lengths can close a triangle.
4. **7.4 Ruler-and-compass constructions** — Use equal radii to force a property, keep construction arcs visible and check the defining relationship.

> **EXPLAIN HOW YOU KNOW**
>
> For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

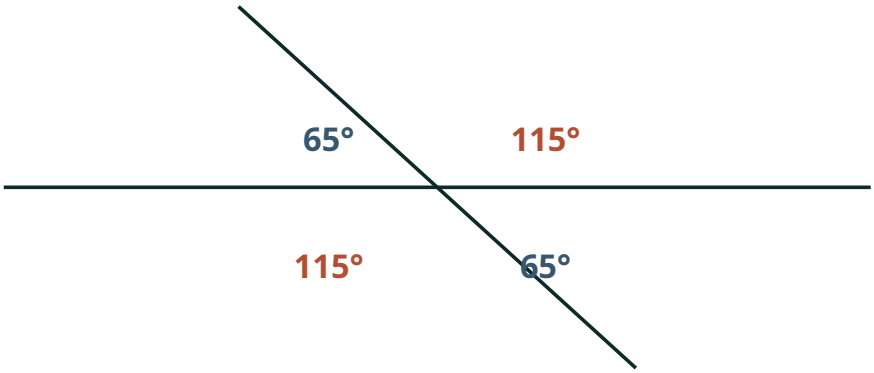
7.1 Lines, angles and angle facts

BIG IDEA

Name the relationship first: straight line, full turn, vertically opposite or right angle.

Angles on a straight line add to 180°; angles around a point add to 360°. Vertically opposite angles are equal. A right angle is 90°.

Classify angles as acute (<90°), right (90°), obtuse (between 90° and 180°) or reflex (between 180° and 360°). A diagram is not always drawn to scale, so use stated facts rather than appearance.



Vertically opposite angles are equal; adjacent angles sum to 180°.

Vertically opposite angles are equal; adjacent angles on a straight line sum to 180°.

WORKED EXAMPLE · FIND A MISSING ANGLE

Two lines cross. One angle is 115°.

1. The vertically opposite angle is 115°.
2. Each adjacent angle is 180°−115° = 65°.

115°, 65°, 115°, 65° around the intersection.

CHAPTER 7 · PRACTICE

7.1 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

Do not assume lines are parallel or angles equal unless marked or stated. “Looks like 90°” is not a reason.

GUIDED PRACTICE · FIRST STEPS

7.1.1 An angle on a straight line is 65°. Find its neighbour. FLUENCY

7.1.2 Two lines cross and one angle is 115°. Find the other three. FLUENCY

7.1.3 Three angles around a point are 90°, 110° and 75°. Find the fourth. REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

7.1.4 Classify 35°, 90°, 125° and 210°. FLUENCY

7.1.5 A ramp makes 12° with the ground. What angle does it make with the vertical? APPLICATION

7.1.6 Can two adjacent angles on a straight line both be obtuse? Explain. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

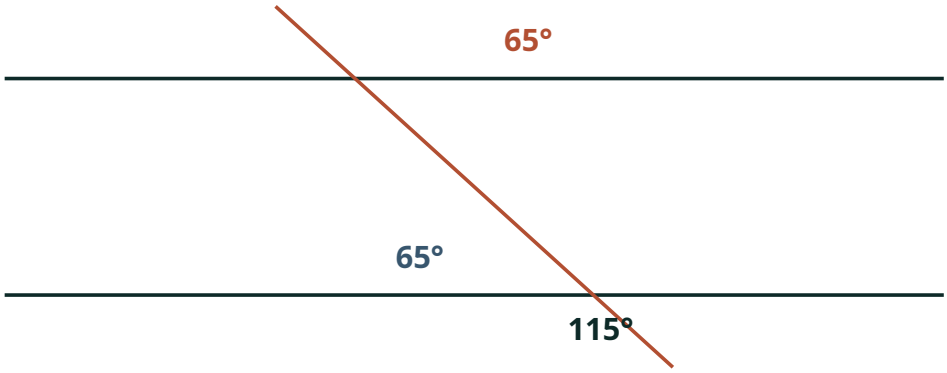
7.2 Parallel lines and triangles

BIG IDEA

Use corresponding, alternate and co-interior angles only when the lines are parallel.

When a transversal crosses parallel lines, corresponding angles are equal, alternate angles are equal, and co-interior angles add to 180°.

Triangle angles sum to 180°. An exterior angle equals the sum of the two opposite interior angles. In an isosceles triangle, the angles opposite the equal sides are equal.



Corresponding and alternate angles are equal; co-interior angles sum to 180°.

With parallel lines: corresponding and alternate angles are 65°; the co-interior angle is 115°.

WORKED EXAMPLE · A ROOF TRUSS

The two base angles are 65° each.

1. Apex = 180−65−65 = 50°.
2. The exterior angle at a base is 180−65 = 115°.

Apex 50°; exterior base angle 115°.

CHAPTER 7 · PRACTICE

7.2 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

The triangle sum is 180°, not 360°. Parallel-line facts need the parallel condition.

GUIDED PRACTICE · FIRST STEPS

7.2.1 A triangle has angles 65° and 65°. Find the third. FLUENCY

7.2.2 Parallel lines are cut by a transversal. One angle is 65°. Find its corresponding, alternate and co-interior angles. FLUENCY

7.2.3 An exterior angle of a triangle is 115° and one opposite interior angle is 65°. Find the other. REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

7.2.4 An isosceles triangle has apex 50°. Find each base angle. APPLICATION

7.2.5 Can a triangle have angles 60°, 70° and 60°? Explain. REASONING

7.2.6 A triangle has angles in the ratio 2:3:4. Find them. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

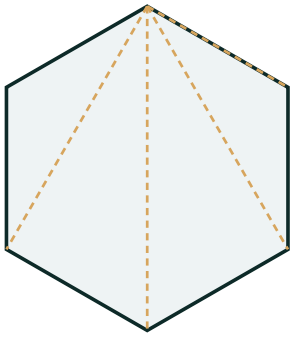
7.3 Polygons and the triangle inequality

BIG IDEA

Split polygons into triangles; test whether three lengths can close a triangle.

An n-sided polygon can be split into n−2 triangles from one vertex, so its interior angle sum is (n−2)×180°. The exterior angles of any convex polygon sum to 360°.

For a regular n-gon, each exterior angle is 360°/n and each interior angle is 180°−360°/n. Three lengths form a non-degenerate triangle exactly when the sum of the two shorter lengths exceeds the longest.



6 sides 4 triangles 720° interior sum

A hexagon splits into four triangles: interior sum 4 × 180° = 720°.

WORKED EXAMPLE · A PENTAGON AND A POSSIBLE TRIANGLE

Find the interior sum of a pentagon; test sides 3, 4, 8.

1. Pentagon: (5−2)×180 = 540°.
2. 3+4 = 7 < 8, so no triangle.

540°; no triangle.

CHAPTER 7 · PRACTICE

7.3 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

Equality gives a straight line, not a triangle. The inequality must be strict.

GUIDED PRACTICE · FIRST STEPS

7.3.1 Find the interior angle sums of a quadrilateral, pentagon and hexagon. FLUENCY

7.3.2 Find each interior and exterior angle of a regular hexagon. FLUENCY

7.3.3 Can sides 3 cm, 4 cm and 8 cm form a triangle? What about 4, 5, 9? REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

7.3.4 A regular polygon has exterior angle 45°. How many sides? APPLICATION

7.3.5 Why is checking only the two shorter sides against the longest enough? REASONING

7.3.6 Two sides are 5 cm and 8 cm. List all whole-number third sides. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

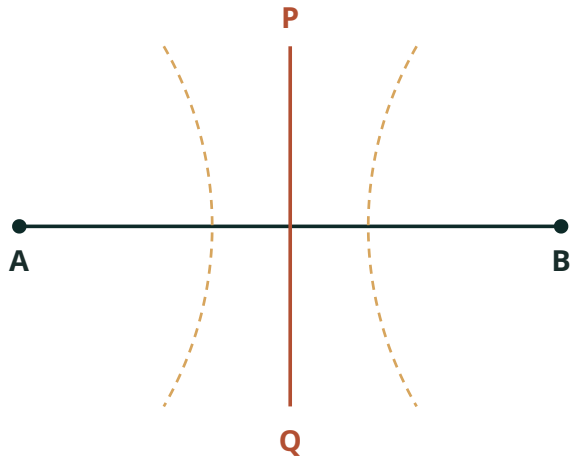
7.4 Ruler-and-compass constructions

BIG IDEA

Use equal radii to force a property, keep construction arcs visible and check the defining relationship.

A perpendicular bisector cuts a segment at its midpoint at 90°. Draw equal-radius arcs from both endpoints, with radius greater than half the segment, then join the arc intersections.

An angle bisector uses an arc from the vertex, then equal arcs from the two arm intersections. For an SSS triangle, draw one side and use the other two lengths as arc radii from its endpoints.



AB = 6 cm; equal radii locate the perpendicular bisector.

Equal radii from A and B locate P and Q. Line PQ is the perpendicular bisector of AB.

WORKED EXAMPLE · CONSTRUCT A 5–6–7 TRIANGLE

Use the longest side as the base.

1. Draw AB = 7 cm.
2. From A draw radius 5 cm; from B draw radius 6 cm. Let the arcs meet at C.
3. Join AC and BC; check 5+6>7.

Triangle ABC has sides 5, 6 and 7 cm.

CHAPTER 7 · PRACTICE

7.4 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

Do not change the compass radius between the paired arcs of a bisector. A protractor drawing is a measurement, not a ruler-and-compass construction.

GUIDED PRACTICE · FIRST STEPS

7.4.1 For a 6 cm segment, what must be true of the compass radius used for a perpendicular bisector? FLUENCY

7.4.2 A 70° angle is bisected by construction. What should each half measure? REASONING

7.4.3 Can a 5 cm, 6 cm, 7 cm triangle be constructed? State the check. APPLICATION

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

7.4.4 Why must construction arcs be left visible? REASONING

7.4.5 Describe how to locate the midpoint of a 6 m base in a 1:20 scale drawing. APPLICATION

7.4.6 A student uses radius 2 cm from both ends of a 6 cm segment. Explain why the perpendicular-bisector construction fails. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 7 · EXPLORATION

The triangle inequality

For each triple, compare the sum of the two shorter sides with the longest. A triangle must close with nonzero area.

> **INVESTIGATION**
> Criterion B. Test triples, identify borderline cases and justify the rule with compass arcs.

SIDES (CM)	POSSIBLE?	REASON
3, 4, 5		
3, 4, 8		
4, 5, 9		
6, 7, 10		
2, 2, 3		

EXPLORATION · THE TRIANGLE INEQUALITY

Criterion B. Test triples, identify borderline cases and justify the rule with compass arcs.

B1 Complete the table for the five triples. INVESTIGATION

B2 State the triangle inequality rule. INVESTIGATION

B3 Test 5, 8, 13; 5, 8, 12; and 7, 7, 14. Identify borderline cases. INVESTIGATION

B4 Two sides are 5 cm and 8 cm. Find every whole-number third side. CHALLENGE

B5 Sides x, x+1 and x+2 are whole numbers. Find the smallest x that makes a triangle, and explain the arcs. CHALLENGE

CHAPTER 7 · CHECKPOINT

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

C1	Angles on a line are x and 115°. Find x.	CHECK
C2	A triangle has angles 50° and 65°. Find the third.	CHECK
C3	Find the interior angle sum of a pentagon.	CHECK
C4	Can sides 2, 2, 3 form a triangle?	CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

CHAPTER 7 · APPLICATION

The roof-truss workshop

A triangular truss has a 6 m base and equal base angles of 65°. A collar beam is parallel to the base.

REAL-LIFE APPLICATION

Criterion D. Simulated workshop design. Use angle facts and a 1:20 scale template before cutting timber.

DESIGN FACT	VALUE
Base	6 m
Base angles	65° each
Scale	1:20

APPLICATION · THE ROOF-TRUSS WORKSHOP

Criterion D. Simulated workshop design. Use angle facts and a 1:20 scale template before cutting timber.

D1 Find the apex angle and classify the triangle. APPLICATION

D2 Find the exterior angle at a base and the base length on the 1:20 drawing. APPLICATION

D3 A second design has base 8 m and rafters 2.5 m each. Can it form a triangle? REASONING

D4 Which construction locates the centre line from the apex to the base? Explain why it passes through the apex. APPLICATION

CHAPTER 7 · ASSESSMENT PRACTICE

The construction card

MYP ASSESSMENT PRACTICE

Criteria A and C. Write a step-by-step card for a 5–6–7 cm triangle and a check that does not rely on appearance.

Your card should be usable by a partner with a ruler and compasses, not a protractor.

ASSESSMENT PRACTICE · THE CONSTRUCTION CARD

Criteria A and C. Write a step-by-step card for a 5–6–7 cm triangle and a check that does not rely on appearance.

S1	Give the construction steps and the inequality check.	ASSESSMENT
S2	Explain how to check the completed triangle.	ASSESSMENT
S3	Write one warning about a common construction error.	ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

CHAPTER 7 · MIXED PRACTICE

Mixed practice 7

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 7

Connect ideas. Give units and reasons where requested.

M1 Find the neighbour of 72° on a straight line. FLUENCY

M2 A triangle has angles 35° and 85°. Find the third. FLUENCY

M3 Find the interior angle sum of an octagon. FLUENCY

M4 A regular polygon has exterior angle 30°. Find its number of sides and interior angle. APPLICATION

M5 Can sides 4, 4, 9 form a triangle? REASONING

M6 Parallel lines give a co-interior angle of 110°. Find its partner. REASONING

M7 A 70° angle is bisected. Find each half. APPLICATION

M8 An isosceles triangle has apex 40°. Find the base angles. CHALLENGE

M9 List whole-number third sides for a triangle with sides 6 and 9. CHALLENGE

M10 A 1:20 drawing has a 30 cm base. Find the real length in metres. APPLICATION

FINAL REFLECTION

Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 7 · ANSWERS AND REASONING

Answers · Ready check and Checkpoint

READY CHECK

R1 **115° and 85°.**
Use straight-line and full-turn totals.

R2 **60.**
Three equal parts of 180.

R3 **Obtuse.**
It lies between 90° and 180°.

R4 **11 > 7.**
Add the two shorter lengths.

R5 **3 cm.**
6÷2.

CHECKPOINT

C1 **65°.**
180−115.

C2 **65°.**
180−115.

C3 **540°.**
(5−2)×180.

C4 **Yes: 2+2>3.**
The strict inequality holds.

CHAPTER 7 · ANSWERS AND REASONING

Answers · 7.1 and 7.2

PRACTICE 7.1 · ANGLE FACTS

7.1.1 **115°.**
180−65.

7.1.2 **65°, 115°, 65°.**
Vertically opposite angles are equal; adjacent angles sum to 180°.

7.1.3 **85°.**
360−(90+110+75).

7.1.4 **Acute, right, obtuse, reflex.**
Compare with 90°, 180° and 360°.

7.1.5 **78°.**
Ground and vertical are perpendicular: 90−12.

7.1.6 **No. Their sum would exceed 180°.**
Each obtuse angle is greater than 90°.

PRACTICE 7.2 · PARALLEL AND TRIANGLE FACTS

7.2.1 **50°.**
180−130.

7.2.2 **65°, 65°, 115°.**
Corresponding and alternate are equal; co-interior sums to 180°.

7.2.3 **50°.**
115−65.

7.2.4 **65°.**
(180−50)÷2.

7.2.5 **No. They sum to 190°, not 180°.**
The angle sum is a necessary condition.

7.2.6 **40°, 60°, 80°.**
Nine parts make 180°, so one part is 20°.

CHAPTER 7 · ANSWERS AND REASONING

Answers · 7.3 and 7.4

PRACTICE 7.3 · POLYGONS AND POSSIBLE TRIANGLES

7.3.1 **360°, 540°, 720°.**
Use (n−2)×180°.

7.3.2 **Interior 120°; exterior 60°.**
360÷6 = 60; 180−60.

7.3.3 **Neither. 3+4<8; 4+5=9 gives a degenerate straight line.**
The two shorter sides must sum to more than the longest.

7.3.4 **8.**
360÷45.

7.3.5 **If a+b>c with c longest, then a+c>b and b+c>a automatically because c is at least as large as the other sides.**
The other two inequalities follow from ordering.

7.3.6 **4, 5, 6, 7, 8, 9, 10, 11, 12 cm.**
Need 3 < x < 13.

PRACTICE 7.4 · CONSTRUCT

7.4.1 **Greater than 3 cm.**
The arcs must meet above and below the segment.

7.4.2 **35°.**
A bisector makes two equal angles.

7.4.3 **Yes: 5+6>7.**
The two shorter sides exceed the longest.

7.4.4 **They record the equal-radius method and allow the construction to be checked.**
The arcs are evidence, not clutter.

7.4.5 **Draw a 30 cm base and construct its perpendicular bisector; it meets the base at 15 cm.**
6 m ÷ 20 = 0.3 m = 30 cm; half is 15 cm.

7.4.6 **The arcs do not meet: 2+2<6. The radius must exceed 3 cm.**
Equal radii are not enough; they must be long enough to intersect.

CHAPTER 7 · ANSWERS AND REASONING

Answers · Exploration

EXPLORATION · THE TRIANGLE INEQUALITY

B1 **Yes, no, no (degenerate), yes, yes.**
Compare the two shorter sums with the longest.

B2 **The sum of the two shorter sides must be strictly greater than the longest side.**
Equality produces a straight line.

B3 **No, yes, no. The first and third are borderline equality cases.**
5+8=13 and 7+7=14.

B4 **4 through 12 cm inclusive.**
$3 < x < 13$.

B5 **x=2. At x=1 the lengths 1,2,3 are degenerate; for x≥2 the two shorter arcs meet.**
x+(x+1)>x+2 gives x>1.

CHAPTER 7 · ANSWERS AND REASONING

Answers · Application and Assessment

APPLICATION · THE ROOF-TRUSS WORKSHOP

D1 **50°; isosceles.**
180−65−65; equal base angles imply equal rafters.

D2 **115°; 30 cm.**
180−65; 600 cm÷20.

D3 **No: 2.5+2.5=5<8.**
The two shorter sides do not reach across the base.

D4 **Construct the perpendicular bisector of the base. In an isosceles triangle the apex is equidistant from the base endpoints, so it lies on that bisector.**
Equal rafters give the equidistance property.

ASSESSMENT PRACTICE · THE CONSTRUCTION CARD

S1 **Draw 7 cm base; arcs of 5 cm and 6 cm from its ends meet at the third vertex; join. Check 5+6>7.**
The arcs transfer the required side lengths.

S2 **Measure all three sides against 5, 6 and 7 cm within drawing tolerance; the angle sum should be 180°.**
A construction is checked by its defining properties.

S3 **Sample: “Do not change the compass radius while transferring a side; keep the arcs visible as evidence.”**
A changed radius changes the required length.

CHAPTER 7 · ANSWERS AND REASONING

Answers · Mixed practice

MIXED PRACTICE 7

M1 **108°.**
180−72.

M2 **60°.**
180−120.

M3 **1080°.**
(8−2)×180.

M4 **12 sides; interior 150°.**
360÷30; 180−30.

M5 **No: 4+4<9.**
The shorter sides do not reach.

M6 **70°.**
180−110.

M7 **35°.**
70÷2.

M8 **70° each.**
(180−40)÷2.

M9 **4 through 14 cm.**
3 < x < 15.

M10 **6 m.**
30×20=600 cm=6 m.